

VisionFusion ■

Week 1 Development Plan

Main Goal

Build the core perception backbone using YOLOv8 and Depth Estimation.

Tasks

- Setup Python environment and GitHub workflow
- Install YOLOv8, OpenCV, PyTorch
- Build webcam inference pipeline
- Add FPS and latency monitoring
- Integrate Depth Anything v2
- Create initial scene dictionary
- Push clean commits to GitHub

Expected Deliverables

- Real-time object detection
- Depth overlays
- Stable webcam pipeline
- Initial demo video

This roadmap is part of the VisionFusion research initiative focused on adaptive multimodal scene understanding using Vision-Language Model fusion.